NETWORK SECURITY

Lecture 1: Principles and Core Concepts

Learning Outcomes

- Define a computer network and network security.
- Explain why network security is important.
- Describe confidentiality, integrity, and availability (CIA Triad).
- Distinguish asset, threat, vulnerability, attack, risk, and security control.
- Compare passive and active attacks.
- Identify administrative, technical, and physical controls.
- Explain defense in depth.

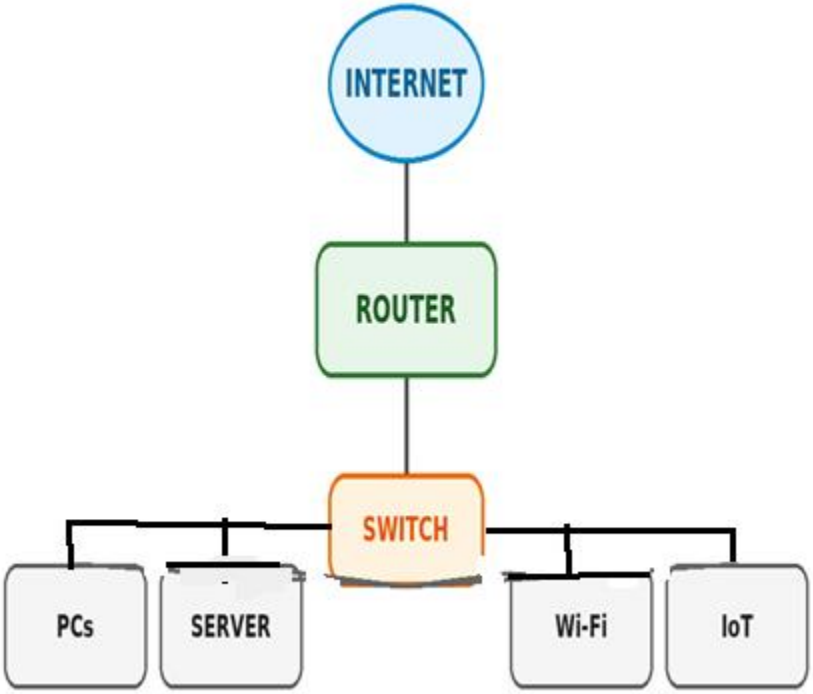
Key idea: Network security protects devices, services, users, and information from unauthorized or harmful activities.

1. Computer Networks and Network Security

What is a Computer Network?

A computer network is a group of connected devices that communicate and exchange information. Common devices include computers, smartphones, servers, switches, routers, wireless access points, cameras, printers, and IoT devices.

Device	Main role
Switch	Connects devices inside a local network
Router	Connects one network to another / Internet
Server	Provides data or services
Wireless AP	Connects wireless devices (provide Wi-Fi)



A simple network: devices communicate through switches and routers.

What is Network Security?

Network security is the protection of network devices, services, users, and information against unauthorized access, misuse, modification, damage, or interruption.

- Protect stored and transmitted data.
- Protect user accounts and passwords.
- Protect servers, applications, devices, and communication channels.

University Example

- Teachers may modify grades only when authorized.
- Students should view only their own grades.
- Attackers should not steal or change records.

- The system should remain available when needed.

Key idea: Connecting devices improves communication, but every connection can also create a security risk.

2. Why Network Security Matters

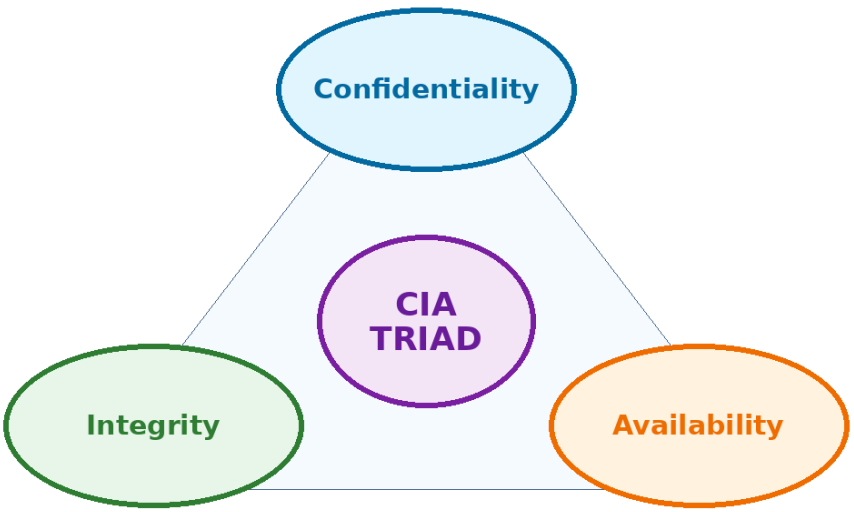
Modern organizations depend on networks for learning, banking, healthcare, government services, business, cloud systems, and mobile communications. A successful security incident can interrupt operations or expose sensitive information.

Environment	Possible incident
University	Stolen teacher password used to change grades
Hospital	Malware prevents access to patient records
Bank	Stolen login information used for unauthorized transactions
Company	Confidential documents copied by an unauthorized employee

Common Consequences

- Loss of confidential information
- Financial loss
- Service interruption
- Loss of important files
- Legal or reputation problems
- Loss of user trust

The Three Main Security Objectives



The CIA Triad is the foundation of information and network security.

3. The CIA Triad

Objective	Meaning	Typical protections
Confidentiality	Only authorized users can read information	Encryption, passwords, access control, VPN
Integrity	Information remains correct and is changed only by authorized users	Hashes, digital signatures, permissions, logging
Availability	Authorized users can access systems and data when needed	Backups, redundancy, UPS, updates, DoS protection

Quick Examples

- Confidentiality violation: an attacker captures an unencrypted username and password.
- Integrity violation: “Transfer $100” is changed to “Transfer $1,000.”
- Availability violation: a server is flooded with traffic and cannot serve legitimate users.

University Examination System

Security objective	Requirement
Confidentiality	Students cannot view exam questions before the exam
Integrity	Students cannot change grades
Availability	The system is accessible during the examination

Key idea: A secure system usually needs all three objectives. Protecting only one is not enough.

4. Authentication, Authorization, and Accountability



Three related concepts that answer three different questions

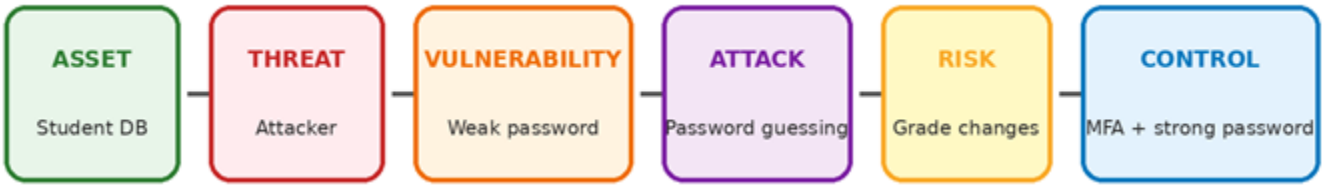
Concept	Question	Example
Authentication	Who are you?	Login using password, fingerprint, smart card, or One-Time Password (OTP).
Authorization	What are you allowed to do?	Student views grades; teacher modifies grades
Accountability	What did you do?	Logs record who changed data, when, and from which device

Important Sequence

1. A user proves identity (authentication).
2. The system checks permissions (authorization).
3. The system records important actions (accountability).

Key idea: Authentication comes first; authorization determines permitted actions after identity is verified.

5. Essential Network Security Terminology



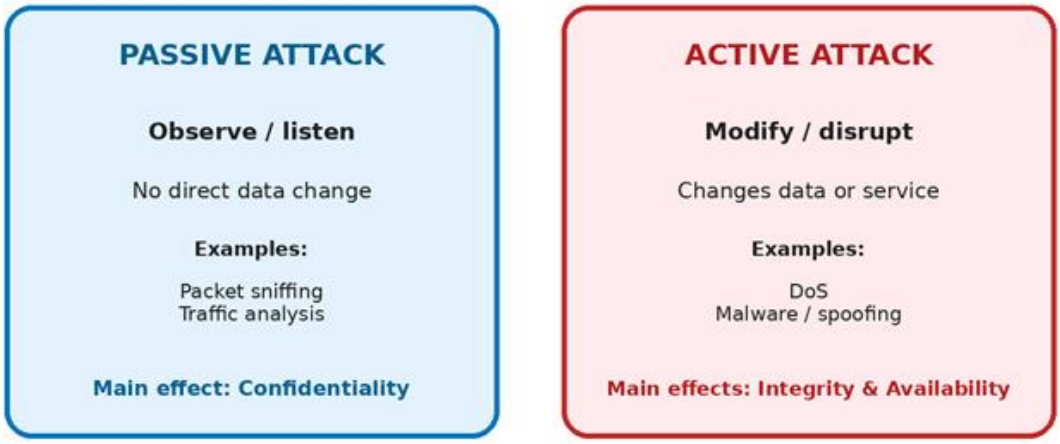
How security terms connect in one simple university example.

Term	Short definition
Asset	Anything valuable that must be protected
Threat	Something that can potentially cause harm
Vulnerability	A weakness in a system, device, configuration, policy, or procedure
Attack	An intentional action that tries to exploit a vulnerability
Risk	The possibility that a threat exploits a vulnerability and causes harm
Security control	A protection used to reduce risk

$$\text{Risk} \approx \text{Likelihood} \times \text{Impact}$$

Key idea: Example: weak administrator password (vulnerability) + attacker (threat) may lead to password guessing (attack) and unauthorized grade changes (risk).

6. Passive and Active Attacks

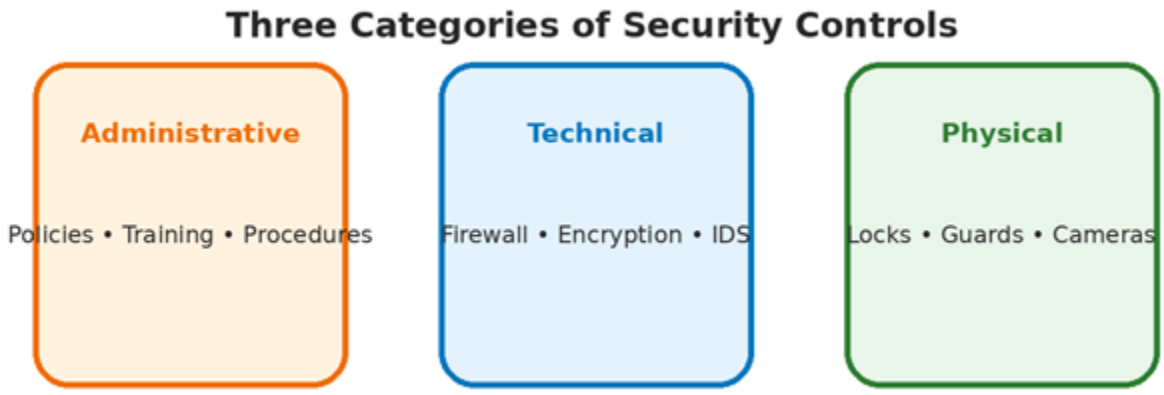


Passive attacks observe; active attacks modify, inject, damage, or interrupt.

Passive attack	Active attack
Observes or collects information	Changes information or system operation
Normally does not modify data	May modify, delete, inject, or disrupt
Often difficult to detect	Often produces visible effects
Mainly affects confidentiality	Mainly affects integrity and availability
Examples: sniffing, traffic analysis	Examples: DoS, malware, spoofing, unauthorized access

Key idea: Encryption is especially useful against passive eavesdropping, while active attacks usually require detection, access control, filtering, and recovery mechanisms as well.

7. Basic Security Controls



Security controls work together; no single category is sufficient by itself.

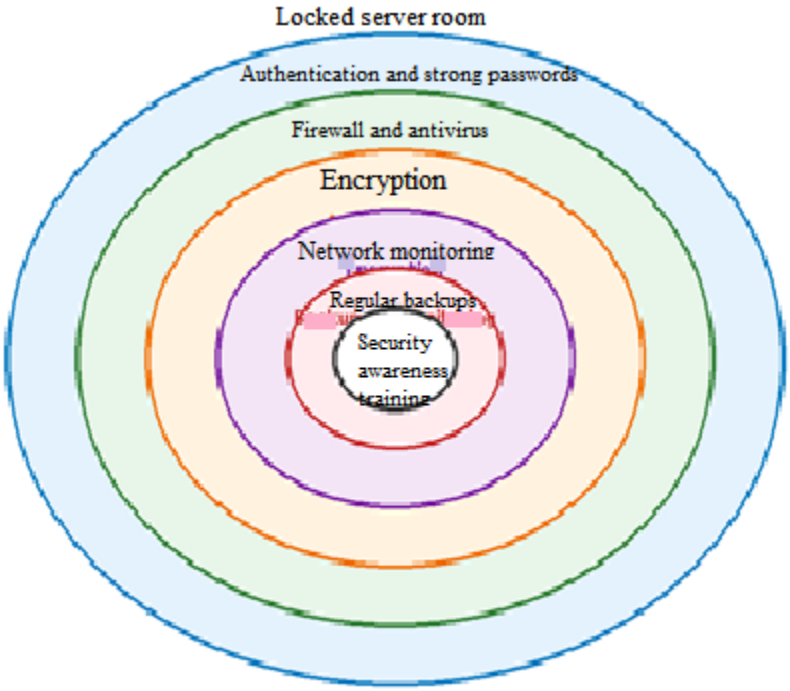
Category	Purpose	Examples
Administrative	Rules and human processes	Security policy, password policy, training, incident procedures
Technical	Hardware or software protection	Firewall, encryption, antivirus, Intrusion Detection System (IDS), authentication, Access Control Lists (ACLs).
Physical	Protect facilities and equipment	Locks, guards, cameras, access cards, fire alarms, backup power

Choosing Controls

Controls should be selected according to the assets being protected, likely threats, known vulnerabilities, and the impact of a possible incident. Strong security usually combines controls from all three categories.

Key idea: A firewall cannot replace user training, and a strong password cannot replace physical protection of a server room.

8. Defense in Depth and a University Case Study



Defense in depth: several protective layers surround important assets.

Defense in Depth

Defense in depth means using several layers of security. If one layer fails, another layer may still reduce the damage.

- Locked server room — Physical protection
- Authentication & strong passwords — Control user access
- Firewall & antivirus — Block threats and malware
- Encryption — Protect sensitive data
- Network monitoring — Detect suspicious activity
- Regular backups — Recover lost data
- Security awareness training — Reduce human errors

Defense in Depth is a security strategy that uses multiple layers of protection to secure a network and its data. These layers may include physical security, strong passwords, firewalls, antivirus software, encryption, network monitoring, backups, and user awareness. If one security layer fails, another layer can still help protect the system. The main idea is that no single security control is enough by itself; stronger security comes from combining several layers together.

Mini Case Study: University Network

Assets	Threats / Vulnerabilities	Useful controls

Student records, exam questions, passwords, research files	Attackers, malware, weak passwords, old software, open Wi-Fi, no backups	Multi-Factor Authentication (MFA), firewall, encryption, updates, antivirus, monitoring, backups, training

9. Ethical Use, Summary, and Review

Ethical Use of Security Tools

- Work only on systems for which you have permission.
- Follow laboratory and university rules.
- Do not access other users' files or capture private information.
- Do not scan external systems without authorization.
- Protect passwords and report security problems to the instructor.

Key idea: Technical ability does not provide legal permission. Practical work must stay inside an authorized environment.

Lecture Summary

- Network security protects devices, users, services, and data.
- The CIA Triad is confidentiality, integrity, and availability.
- Authentication verifies identity; authorization determines permissions.
- Risk connects assets, threats, vulnerabilities, attacks, and impact.
- Passive attacks observe; active attacks change or interrupt.
- Security controls may be administrative, technical, or physical.
- Defense in depth uses multiple security layers.

Review Questions

1. What is network security?
2. What are the three elements of the CIA Triad?
3. What is the difference between authentication and authorization?
4. Distinguish threat, vulnerability, attack, and risk.
5. What is the difference between passive and active attacks?
6. Name the three categories of security controls.
7. What does defense in depth mean?

References: W. Stallings, Network Security Essentials; Cisco Networking Academy materials; NIST cybersecurity terminology.